Haris Khan

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Achievements

Ranked in the **Top 5% globally** (444th / 9000+) and **8th nationally** in **Meta Hacker Cup 2025** (View Here).

Selected as a **Judge** for **Student HackPad 2025**, an international hackathon (800+ participants, $14K+ prizes) (View Here).

Led team at **Harvard's CS50x Puzzle Day 2025**, solving **8/9 puzzles** among global participants (View Here).

Ranked 3rd at **HackHub by SparkHub 2025**, an international hackathon, for an AI-powered exam planner (View Here).

Ranked in the **Top 10% globally** at the **International Computer Science Competition (ICSC)** (View Here).

Ranked **177th out of 355** in the **MIT Informatics Contest** (View Here).

Achieved a **TOEFL score of 91**, equivalent to **IELTS Band 7.0** (View Here).

Education

University of Peshawar 2020 – 2024

Bachelor of Computer Science – CGPA: 3.37/4.00 *Peshawar, Pakistan*

- **Final Year Project:** Outdoor Object Detection System for the Visually Impaired (View Here)
- **Tools:** Python, YOLOv8, OpenCV, Google Text-to-Speech
- Developed a YOLOv8-based real-time system to detect common outdoor obstacles using a custom dataset, achieving **0.87 precision** and **0.82 F1-score**.
- Integrated text-to-speech alerts to provide verbal navigation guidance for visually impaired users.
- **Impact:** Enhanced mobility independence through real-time obstacle detection and audio feedback, helping reduce navigation-related risks.

Publications

I. Ahmad, **M. Haris Khan**, A. Khan, D. Ramli, M. Asim, M. Isha, and A. Khan, *"Fruits Plant Leaf Diseases Detection using YOLOv12 Model with Flash Attention Mechanism and R-ELAN Block,"* (submitted to *Smart Agricultural Technology*.)

Experience

The University of Agriculture, Peshawar Jan 2025 – Present

Research Assistant *Peshawar, Pakistan*

- **Supervisor:** Dr. Abdullah, Associate Professor.
- Contributed to two research projects involving deep learning and computer vision models.
- Cleaned, preprocessed, and organized large-scale datasets to support reliable model training.
- Trained and fine-tuned deep learning models, including YOLO-based architectures.
- Assisted in research writing, methodology development, and experimental result analysis.

Projects

Lung Cancer Detection using Hybrid Deep Learning and Explainable AI (View Here)

- Developed a hybrid CT scan classification model using InceptionV3, attention mechanisms, and GLCM/LBP features for multi-class detection.
- Integrated Grad-CAM for explainable AI to highlight clinically relevant regions influencing predictions.
- Achieved **98.9% accuracy** and **0.997 ROC-AUC** with a complete preprocessing and evaluation pipeline.

Heart Sound Classification using Lightweight Deep Learning (View Here)

- Built a deep learning system to classify normal vs abnormal PCG signals for biomedical analysis.
- Designed and benchmarked lightweight models including Tiny 2D CNN, 1D CNN, and YAMNet transfer learning.
- Achieved **92.44% accuracy** with optimized F1-score, inference speed, and energy efficiency.

EfficientUNetViT-Based Breast Tumor Segmentation (View Here)

- Developed a deep learning pipeline for breast tumor segmentation from ultrasound images.
- Evaluated models including EfficientUNetViT, DeepLabV3+, MobileNetV2-UNet, and ensemble approaches.
- Achieved **95.9% accuracy**, **0.718 Dice score**, and **0.647 IoU** for improved tumor localization.

Anomalous Behavior Detection in Video Surveillance (View Here)

- Implemented a BiLSTM-based video anomaly detection model inspired by the IEEE paper *"A New Approach to Detect Anomalous Behavior in Video Surveillance"*.
- Built a pipeline with frame grouping, ResNet-50 feature extraction, and BiLSTM temporal modeling.
- Achieved **80% accuracy**, **0.78 F1-score**, and **0.87 ROC-AUC** with visual performance analysis.

International Hackathons

NASA Space Apps Challenge 2025 — By NASA

Exoplanet Hunter *(View Here)*

Role: Team Lead & Machine Learning Developer

- Developed an ML model using NASA's TESS mission dataset to classify exoplanet candidates.
- Achieved **89% accuracy** using the XGBoost algorithm.
- Built a **Streamlit web app** for real-time exoplanet prediction and visualization.

Co-Creating with GPT-5 — By Lablab.ai

Eco Mentor *(View Here)*

Role: Full-Stack Developer

- Built a GPT-5-powered sustainability mentor for kids with interactive chat, quizzes, and real-life eco challenges.
- Integrated gamification with points, badges, and progress tracking to improve user engagement.
- **Tech:** FastAPI, React, GPT-5 API, Railway, Vercel.

HackHub 2025 — By SparkHub

Exam Companion – Exam Preparation Assistant *(View Here)*

Role: Full-Stack Developer

- Built a web app to help students plan, track, and manage exam study schedules.
- Developed interactive dashboards, schedule export options, and a responsive interface.

Volunteer Work

ICodeGuru — Remote – Silicon Valley, USA

Trainer & Moderator — *Feb 2025 – Present*

- Taught core data structures and LeetCode problem-solving patterns to 50+ underprivileged students over six weeks. (View Here).
- Led a 5-week **IELTS Preparation Course**, delivering high-impact strategies, real-time feedback, and speaking/writing improvement (View Here).

Alkhidmat Foundation

Relief Distribution Volunteer — *2023 – Present*

- Assisted in meal distribution drives for underprivileged families across remote and low-income communities.
- Supported Ramadan relief efforts through coordination, teamwork, and community service delivery.

Workshops

Transformers Architecture: From Attention to LLMs Workshop (View Here)

- Delivered a workshop on the evolution of NLP models leading to Transformer architecture and LLMs.
- Explained core concepts including attention mechanism, self-attention, and encoder–decoder architecture.
- Conducted hands-on sessions in Google Colab demonstrating Transformer applications such as sentiment analysis.

Applied Computer Vision with YOLO: Detection & Segmentation (View Here)

- Delivered sessions on computer vision fundamentals and YOLO evolution for real-time detection.
- Guided participants in building, annotating, and training custom datasets.
- Showcased YOLO applications in healthcare, autonomous systems, and robotics.

Exploring NASA Space Apps Challenge 2025 (View Here)

- Introduced participants to NASA's Space Apps Challenge Hackathon and open data resources.
- Explained key 2025 challenges, including data-driven climate, ocean, and space exploration projects.
- Guided students on team formation, project selection, and hackathon preparation.

Leadership

HEC Generative AI Training Hackathon (View Here)

Judge and Mentor

- Mentored teams in idea development and technical implementation, improving the quality of Generative AI projects.
- Evaluated projects and presentations to select finalists through fair and rigorous technical assessment.

Student HackPad 2025 (View Here)

Judge

- Evaluated 100+ student-led educational projects for innovation and learning impact.
- Provided structured feedback on project design and implementation.

HealTech Innovators Hackathon 2025 (View Here)

Judge

- Evaluated AI-driven healthcare projects for impact and technical quality.
- Assessed project feasibility and real-world healthcare applicability.

Certifications

Artificial Intelligence Bootcamp – Issued by Atomcamp (View Here).

Data Science Bootcamp – Issued by Atomcamp (View Here).

AI and Machine Learning – Issued by Provincial Government of Khyber Pakhtunkhwa (KPITB) (View Here).

Technical Skills

Programming Languages: Python, PHP, C++
Databases: MySQL, SQLite
Frameworks & Libraries: YOLOv8/v9/v10, Streamlit, Flask, OpenCV, Scikit-learn, PyTorch
Tools & Platforms: Git, GitHub, Google Colab, n8n, Hugging Face Spaces
APIs & Services: OpenAI API, Groq API, Claude API, Qloo API